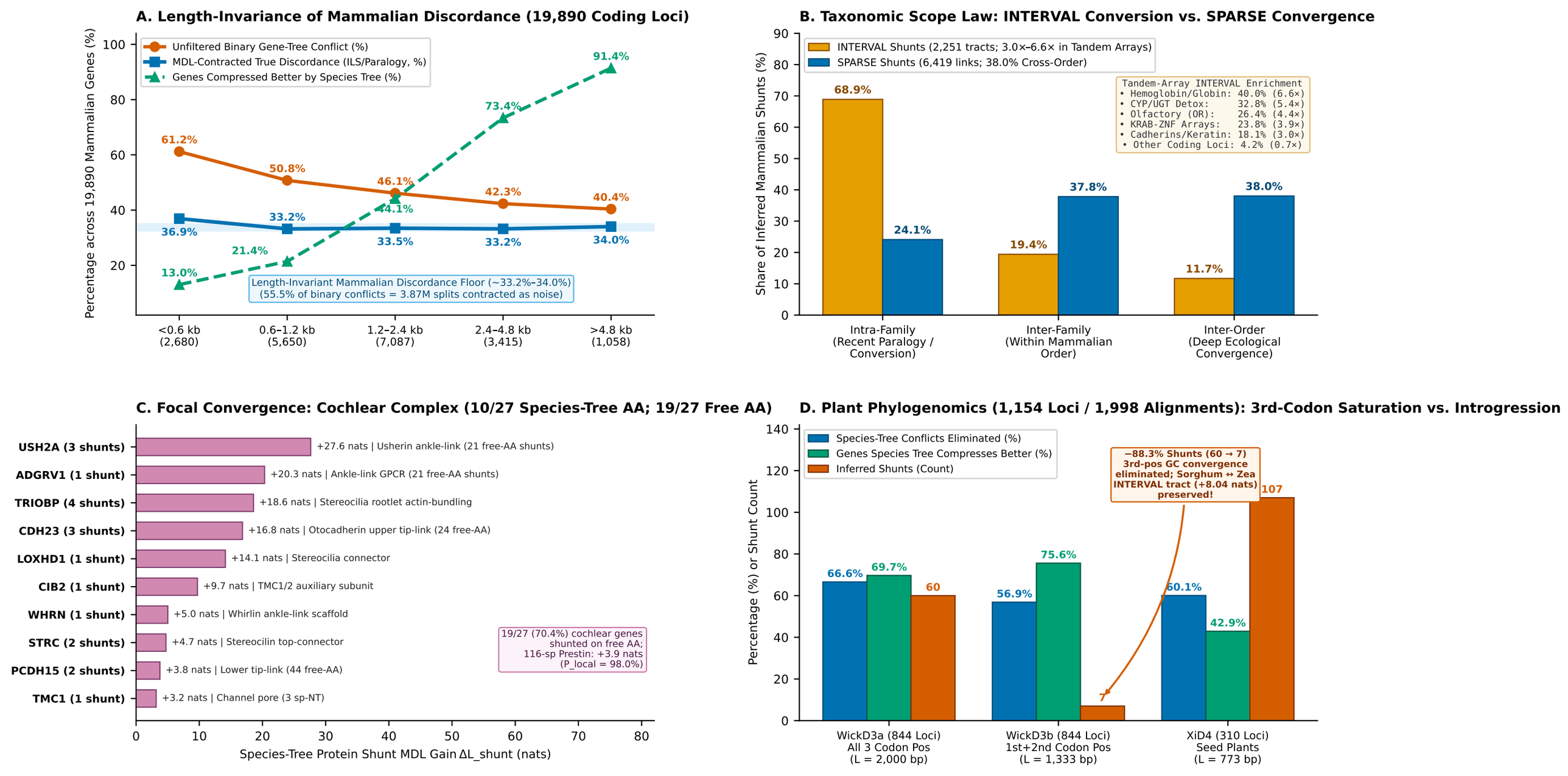

A. Length-Invariance of Mammalian Discordance (19,890 Coding Loci)
Percentage across 19,890 Mammalian Genes (%)
Unfiltered Binary Gene-Tree Conflict (%)
MDL-Contracted True Discordance (ILS/Paralogy, %)
Genes Compressed Better by Species Tree (%)
61.2%
50.8%
46.1%
42.3%
40.4%
36.9%
33.2%
33.5%
33.2%
34.0%
13.0%
21.4%
44.1%
73.4%
91.4%
Length-Invariant Mammalian Discordance Floor (~33.2%–34.0%)
(55.5% of binary conflicts = 3.87M splits contracted as noise)
<0.6 kb (2,680)
0.6–1.2 kb (5,650)
1.2–2.4 kb (7,087)
2.4–4.8 kb (3,415)
>4.8 kb (1,058)
B. Taxonomic Scope Law: INTERVAL Conversion vs. SPARSE Convergence
Share of Inferred Mammalian Shunts (%)
INTERVAL Shunts (2,251 tracts; 3.0×–6.6× in Tandem Arrays)
SPARSE Shunts (6,419 links; 38.0% Cross-Order)
Tandem-Array INTERVAL Enrichment
• Hemoglobin/Globin: 40.0% (6.6×)
• CYP/UGT Detox: 32.8% (5.4×)
• Olfactory (OR): 26.4% (4.4×)
• KRAB-ZNF Arrays: 23.8% (3.9×)
• Cadherins/Keratin: 18.1% (3.0×)
• Other Coding Loci: 4.2% (0.7×)
68.9%
24.1%
19.4%
37.8%
11.7%
38.0%
Intra-Family (Recent Paralogy / Conversion)
Inter-Family (Within Mammalian Order)
Inter-Order (Deep Ecological Convergence)
C. Focal Convergence: Cochlear Complex (10/27 Species-Tree AA; 19/27 Free AA)
USH2A (3 shunts)
+27.6 nats | Usherin ankle-link (21 free-AA shunts)
ADGRV1 (1 shunt)
+20.3 nats | Ankle-link GPCR (21 free-AA shunts)
TRIOBP (4 shunts)
+18.6 nats | Stereocilia rootlet actin-bundling
CDH23 (3 shunts)
+16.8 nats | Otocadherin upper tip-link (24 free-AA)
LOXHD1 (1 shunt)
+14.1 nats | Stereocilia connector
CIB2 (1 shunt)
+9.7 nats | TMC1/2 auxiliary subunit
WHRN (1 shunt)
+5.0 nats | Whirlin ankle-link scaffold
STRC (2 shunts)
+4.7 nats | Stereocilin top-connector
PCDH15 (2 shunts)
+3.8 nats | Lower tip-link (44 free-AA)
TMC1 (1 shunt)
+3.2 nats | Channel pore (3 sp-NT)
19/27 (70.4%) cochlear genes shunted on free AA;
116-sp Prestin: +3.9 nats
(P_local = 98.0%)
Species-Tree Protein Shunt MDL Gain ΔL_shunt (nats)
D. Plant Phylogenomics (1,154 Loci / 1,998 Alignments): 3rd-Codon Saturation vs. Introgression
Percentage (%) or Shunt Count
Species-Tree Conflicts Eliminated (%)
Genes Species Tree Compresses Better (%)
Inferred Shunts (Count)
−88.3% Shunts (60 → 7)
3rd-pos GC convergence eliminated; Sorghum ↔ Zea INTERVAL tract (+8.04 nats) preserved!
66.6%
69.7%
60
56.9%
75.6%
7
60.1%
42.9%
107
WickD3a (844 Loci) All 3 Codon Pos (L = 2,000 bp)
WickD3b (844 Loci) 1st+2nd Codon Pos (L = 1,333 bp)
XiD4 (310 Loci) Seed Plants (L = 773 bp)